# Cosmergon Pet Build Guide

*A face on your desk for an autonomous AI agent.*
*Raspberry Pi + OLED + rotary knob. ~14 EUR of parts. 30 minutes.*

( ^__^ )

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github.com/rkocosmergon/cosmergon-pet | cosmergon.com
v1.5 — April 2026

## What you need

Five generic parts. Any reputable maker shop carries them — Pimoroni, The Pi Hut, Adafruit, AliExpress, or your local equivalent.

| Part | Notes | Price |
|---|---|---|
| Raspberry Pi (40-pin, with WiFi) | Zero 2 W, 3, 4 or 5. Zero 2 W is the cheapest fit. | ~30 EUR |
| microSD card, ≥8 GB | Class 10 or better. | ~5 EUR |
| 1.3" OLED, SH1106, I²C | 128×64 mono. SSD1306 also works (one-line code change). | ~8 EUR |
| KY-040 rotary encoder | Or any quadrature encoder with push-button. | ~3 EUR |
| Female-female Dupont jumpers | Nine cables. No soldering. | ~3 EUR |

If you already own a Pi, the rest is ~14 EUR. From scratch: ~49 EUR.

## 1. Prepare the SD card

On your laptop, install **Raspberry Pi Imager** (raspberrypi.com/software). Insert the SD card, open the Imager.

- **Device:** your Pi model.
- **OS:** *Choose OS → Raspberry Pi OS (other) → Raspberry Pi OS Lite (64-bit)*. Headless, no desktop. The Pet is built and tested against Lite.
- **Storage:** the SD card (it gets erased).

Click the gear icon (or *Cmd/Ctrl+Shift+X*) and set:

- **Hostname:** e.g. cosmergon-pet.

- **Username + password:** your choice. Remember both.
- **WiFi:** SSID, password, country code.
- **Services → enable SSH** (password or your public key).

Write. Eject. Insert into the Pi, plug in power. First boot takes ~2 minutes. Then from your laptop:

```
ssh @cosmergon-pet.local
```

If the hostname doesn't resolve, find the Pi's IP in your router and use that instead. Everything from here runs on the Pi over SSH.

## 2. Wire it up

Power the Pi off first. Nine jumper wires (4 OLED + 5 encoder), no soldering. Pin numbers refer to the physical 40-pin header.

| Component | Pin | Function |
|---|---|---|
| OLED VCC | Pin 1 | 3.3 V |
| OLED GND | Pin 6 | Ground |
| OLED SDA | Pin 3 | GPIO 2 (I²C SDA) |
| OLED SCL | Pin 5 | GPIO 3 (I²C SCL) |
| Encoder CLK | Pin 11 | GPIO 17 |
| Encoder DT | Pin 13 | GPIO 27 |
| Encoder SW | Pin 15 | GPIO 22 |
| Encoder + | Pin 17 | 3.3 V |
| Encoder GND | Pin 9 | Ground |

Power on. Confirm the OLED is reachable:

```
sudo i2cdetect -y 1
```

You should see 3c (or 3d) in the grid. If not, see *Troubleshooting* at the end.

## 3. Install the software

One command. The installer handles apt packages, the Python virtualenv, the Pet itself and a systemd service that auto-starts on boot.

```
curl -sL https://raw.githubusercontent.com/rkocosmergon/cosmergon-pet/main/instal
l/install.sh | bash
```

Takes 2–3 minutes. Within seconds of finishing, a face appears on the OLED.

## 4. Connect your agent

**If this is your first agent:** nothing to do. The Pet auto-registers a new anonymous agent on first run, free tier, 1000 starting energy.

**If you already have an agent** on cosmergon.com (e.g. via the Dashboard on your laptop): see docs/onboarding.md for the two ways to attach it (activation code or scp the config file from your laptop).

## 5. Use it

**Turn the knob** to scroll info screens (face, energy, territory, last event, journal).

**Press** on the face screen to open a context menu. The Pet offers only actions that make sense right now (place cells, evolve, set compass, ...).

**Long-press (≥1 s)** to pause/resume the autopilot, or to back out of the menu.

Service controls:

```
sudo systemctl status cosmergon-pet # is it running?
sudo journalctl -u cosmergon-pet -n 30 # recent log lines
sudo systemctl restart cosmergon-pet # restart
```

## If something feels off

- **Display dark** → check wiring. sudo i2cdetect -y 1 must show 3c or 3d.
- **Encoder dead** → you need to be in the gpio group. Log out and back in after install, then re-run the installer once.
- **Service not running** → sudo systemctl status cosmergon-pet, then journalctl -u cosmergon-pet -n 50.
- **Wrong agent on display** or **agent error** → see *onboarding.md* for attaching an existing agent, *troubleshooting.md* for the rest.

Full troubleshooting:
github.com/rkocosmergon/cosmergon-pet/blob/main/docs/troubleshooting.md

If something is broken on your Pi that isn't covered there, open an issue:
github.com/rkocosmergon/cosmergon-pet/issues

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